

Math 475-01 (Combinatorics and Graph Theory)

Fall Semester 2026

Lecture: TuTh 8:30 – 9:45am Math & Psychology 401

Instructor: Dr. Kalman M. Nanes

Office: Math/Psych 439, phone 410-455-2439

Office hours: M 10am – 12pm, Tu 10 - 11am, and F 10am – 12pm, or by appointment.

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Textbook: Morris, J. *Combinatorics, 2022 Edition*. University of Lethbridge, 2016. This is an open-source publication that can be found for free at www.cs.uleth.ca/~morris/Combinatorics/Combinatorics.html.

I also recommend *A walk through combinatorics* by Miklós Bóna.

Grading: The course will be graded at least as generously as the framework:

- A: 90-100%
- B: 80-89%
- C: 65-79%
- D: 55-64%

Your grade will be calculated as follows:

Homework and Board Presentations (approx. weekly)	15%
Exams (2)	25% each
Final Exam	35%

Learning Objectives: This course will act as a survey covering several bedrock topics in upper level discrete mathematics. We will begin by looking at Combinatorics - the art of counting without counting. We will focus on counting structures, including Pigeonhole Principle, Generating Functions, and Inclusion-Exclusion and Recurrence. The second half of the term will concern Graph Theory, including Adjacency Matrices, Trees, Connectivity, Planarity, Colorings, and ending with a look at Ramsey Theory.

Attendance and Behavior: Attendance is crucial to success in this class. It is also a crucial part of professional behavior. I expect everybody to show up to class on time, every day. **You are responsible for making up any missed material.** I reserve the right to not accept late homework assignments. Make-up quizzes and exams will be given **at my discretion**, in general only in situations where there is a conflict with a **sanctioned university activity or religious holiday** of which I have been notified **at least a week in advance**. Such activities may include things like a course conflict or an athletic event. I will excuse a missed assessment due to illness only if I am notified **before the assessment**, and only if you furnish a **signed doctor's note** at the earliest available opportunity.

As we are all adults here, I expect us to all act like adults and treat each other professionally. This means that everybody should treat each other with respect in this class, including as it pertains to use of technology in the classroom. Therefore, **there will be a ban on cell phones, laptops, tablets, earphones, and other mobile devices in class, except in cases where they are being directly used for classroom purposes.**

Homework: As we progress through the textbook, I recommend completing book exercises as practice. I will also post problem sets for you to hand in most weeks on Monday evenings. Each homework assignment will be graded out of 25 points. Your lowest three scores will be dropped. The remaining total will be weighted as 15% of your overall grade in the course.

Homework expectations are as follows. First, you should view your homework as a professional product. Homework will be submitted online through Blackboard. It should be submitted in a standard recognized file format. It should be neat, legible, and well-organized. You may type up your homework in L^AT_EX if you wish, but do not let it adversely affect the quality of your writing. You should use appropriate mathematical logic and proof methods. You don't need to write **formally** with tortured language and lots of big words; I only expect you to write **clearly** and **rigorously**.

I heavily suggest that you form study groups to work on the practice problems and homework assignments. Working with others is the best way to learn mathematics (or really, any subject). Be aware, however, that you do need to complete your own write-ups for homework assignments. If I receive identical write-ups from different people, I will consider that to be plagiarism, and will prosecute you accordingly. To review: discussing concepts and strategies together is fantastic. Checking your work together or arguing your proofs with each other is great. Writing your homework together word-for-word is very bad. In addition, **homework must represent your own efforts**. Please legitimately try the problems before turning to ChatGPT*. Make sure that even if you lean on AI and other tools, that you put the arguments in your own words, using your own brain at some point in the process.

*: Let's be really clear about ChatGPT and other AI assistants for a moment. Moral issues aside, these programs are getting better at doing mathematics all the time. A year ago, I could truthfully say they couldn't really be trusted for accuracy. Now, I can't really be sure. What you need to know is that there is a responsible way to use AI, and an irresponsible way. Asking AI to just write solutions for you is a bad idea - you won't be developing any "mathematical muscle" that way. An alternate approach, however, might be to ask an AI for hints on a problem without solving it for you, or to explain a difficult topic for you. You might even try running your homework attempts by an AI to see if it can spot any holes in your arguments, or asking an AI to come up with practice material before your quizzes and exams. The thing to remember is that there are no shortcuts to learning mathematics. AI can be a useful tool, but it can't learn for you.

Board Work: An important aspect of this course is learning how to assess mathematical writing and to bounce ideas off of each other. Each Tuesday, we will begin class by selecting volunteers to present their work on one of the homework problems to the class. Then, the class will analyze the work, determine if it is correct, and help amend it if it is not correct. Board work will be graded based on participation and good-faith effort. Each student will be required to present at least one problem during the term, for 25% of your homework grade. Preference will be given to volunteers who have not yet presented a problem during the term.

Landmark Proofs: Certain theorems throughout the term have proofs that demonstrate important ideas in mathematical development. We will call these Landmark Proofs and will spend a certain amount of focus on them during lectures. An important part of each quiz and exam will be to discuss the logic of given Landmark Proofs, so you will want to spend some time understanding these proof methods. I will do my best to make clear which proofs are candidates for this requirement, and to make clear my expectations as we move through the material.

Exams: There will be two in-class midterm exams given during the term, according to the schedule below. In-class exams will be closed-book and closed-notes. Each exam will be graded out of 100 points and will be worth 25% of your overall grade. For policies on make-up exams, see the attendance policy, above.

Final Exam: The final exam will be an in-class exam, held **Thursday, December 17th, from 8:00 - 10:00am, in our usual classroom**. The final exam, like our other exams, will be closed-book and closed-notes. The final exam will be graded out of 200 points and will be worth 35% of your overall grade.

Academic Integrity: By enrolling in this course, each student assumes the responsibilities of an active participant in UMBC's scholarly community in which everyone's academic work and behavior are held to the highest standards of honesty. Cheating, fabrication, plagiarism, and helping others to commit these acts are all forms of academic dishonesty, and they are wrong. Academic misconduct could result in disciplinary action that may include but is not limited to suspension or dismissal. To read the full Student Academic Conduct Policy, consult the UMBC Student Handbook, the Faculty Handbook, or the UMBC Policies section of the UMBC Directory. <http://www.umbc.edu/provost/integrity/faculty.html>

Academic integrity is a core value at UMBC and I expect that value to be reflected in your coursework. You must cite any outside sources you use for homework. You may not use any outside sources for exams or quizzes. You may be asked to have a one-on-one meeting with me to discuss course material. If your responses are substantially different from work you've submitted, you may face an allegation of academic dishonesty, which can result in a zero for the assignment in question and a report of academic misconduct.

Accessibility and Disability Accommodations, Guidance and Resources: Accommodations for students with disabilities are provided for all students with a qualified disability under the Americans with Disabilities Act (ADA & ADAAA) and Section 504 of the Rehabilitation Act who request and are eligible for accommodations. The Office of Student Disability Services (SDS) is the UMBC department designated to coordinate accommodations that creates equal access for students when barriers to participation exist in University courses, programs, or activities.

If you have a documented disability and need to request academic accommodations in your courses, please refer to the SDS website at sds.umbc.edu for registration information and office procedures.

SDS email: disAbility@umbc.edu

SDS phone: 410-455-2459

If you will be using SDS approved accommodations in this class, please contact the instructor to discuss implementation of the accommodations. During remote instruction requirements due to COVID, communication and flexibility will be essential for success.

Campus Safety: The following sections refer to campus policy on harrassment, violence, and other issues. All policies and forms can be found at ecr.umbc.edu.

Sexual Assault, Sexual Harassment, and Gender Based Violence and Discrimination UMBC Policy in addition to federal and state law (to include Title IX) prohibits discrimination and harassment on the basis of sex, sexual orientation, and gender identity in University programs and activities. Any student who is impacted by sexual harassment, sexual assault, domestic violence, dating violence, stalking, sexual exploitation, gender discrimination, pregnancy discrimination, gender-based harassment, or related retaliation should contact the University's Title IX Coordinator to make a report and/or access support and resources. The Title IX Coordinator can be reached at titleixcoordinator@umbc.edu or 410-455-1717.

You can access support and resources even if you do not want to take any further action. You will not be forced to file a formal complaint or police report. Please be aware that the University may take action on its own if essential to protect the safety of the community.

If you are interested in making a report, please use the Online Reporting/Referral Form at https://umbc-advocate.symplicity.com/titleix_report/index.php/pid364290?. Please note that, if you report anonymously, the University's ability to respond will be limited.

Notice that Faculty and Teaching Assistants are Responsible Employees with Mandatory Reporting Obligations

All faculty members and teaching assistants are considered Responsible Employees, per UMBC's Policy on Sexual Misconduct, Sexual Harassment, and Gender Discrimination. Faculty and teaching assistants therefore required to report all known information regarding alleged conduct that may be a violation of the Policy to the Title IX Coordinator, even if a student discloses an experience that occurred before attending UMBC and/or an incident that only involves people not affiliated with UMBC. Reports are required regardless of the amount of detail provided and even in instances where support has already been offered or received.

While faculty members want to encourage you to share information related to your life experiences through discussion and written work, students should understand that faculty are required to report past and present sexual harassment, sexual assault, domestic and dating violence, stalking, and gender discrimination that is shared with them to the Title IX Coordinator so that the University can inform students of their rights, resources, and support. While you are encouraged to do so, you are not obligated to respond to outreach conducted as a result of a report to the Title IX Coordinator.

If you need to speak with someone in confidence, who does not have an obligation to report to the Title IX Coordinator, UMBC has a number of Confidential Resources available to support you:

Retriever Integrated Health (Main Campus): health.umbc.edu 410-455-2472; Monday – Friday 8:30 a.m. – 5 p.m.; For After-Hours Support, Call 988.

Center for Counseling and Well-Being (Shady Grove Campus): 301-738-6273; Monday-Thursday 10:00a.m. – 7:00 p.m. and Friday 10:00 a.m. – 2:00 p.m. (virtual) Online Appointment Request Form <https://shadygrove.titaniumhwc.com/>

Pastoral Counseling via The Gathering Space for Spiritual Well-Being: 410-455-6795; i3b@umbc.edu; Monday – Friday 8:00 a.m. – 10:00 p.m.

Other Resources

Women's Center (open to students of all genders): <https://womenscenter.umbc.edu/> 410-455-2714; womenscenter@umbc.edu; Monday – Thursday 9:30 a.m. – 5:00 p.m. and Friday 10:00 a.m. – 4 p.m.

Shady Grove Student Resources (<https://ecr.umbc.edu/shady-grove-title-ix-resources/>), Maryland Resources (<https://ecr.umbc.edu/maryland-resources/>), and National Resources (<https://ecr.umbc.edu/national-resources/>)

Child Abuse and Neglect

Please note that Maryland law and UMBC policy require that faculty report all disclosures or suspicions of child abuse or neglect to the Department of Social Services and/or the police even if the person who experienced the abuse or neglect is now over 18.

Pregnant and Parenting Students

UMBC's Policy on Sexual Misconduct, Sexual Harassment and Gender Discrimination expressly prohibits all forms of discrimination and harassment on the basis of sex, including pregnancy. Resources for pregnant, parenting and breastfeeding students are available through the University's Office of Equity and Civil Rights. Pregnant and parenting students are encouraged to contact the Title IX Coordinator to discuss plans and ensure ongoing access to their academic program with respect to a leave of absence – returning following leave, or any other accommodation that may be needed related to pregnancy, childbirth, adoption, breastfeeding, and/or the early months of parenting.

In addition, students who are pregnant and have an impairment related to their pregnancy that qualifies as disability under the ADA may be entitled to accommodations through the Office of Student Disability Services.

Religious Observances & Accommodations

UMBC Policy provides that students should not be penalized because of observances of their religious beliefs, and that students shall be given an opportunity, whenever feasible, to make up within a reasonable time any academic assignment that is missed due to individual participation in religious observances. It is the responsibility of the student to inform the instructor of any intended absences or requested modifications for religious observances in advance, and as early as possible. For questions or guidance regarding religious observances and accommodations, please contact the Office of Equity and Civil Rights at ecr@umbc.edu.

Hate, Bias, Discrimination and Harassment

UMBC values safety, cultural and ethnic diversity, social responsibility, lifelong learning, equity, and civic engagement.

Consistent with these principles, UMBC Policy prohibits discrimination and harassment in its educational programs and activities or with respect to employment terms and conditions based on race, creed, color, religion, sex, gender, pregnancy, ancestry, age, gender identity or expression, national origin, veterans status, marital status, sexual orientation, physical or mental disability, or genetic information.

Students (and faculty and staff) who experience discrimination, harassment, hate, or bias based upon a protected status or who have such matters reported to them should use the online reporting/referral form (https://umbc-advocate.symplcity.com/titleix_report/index.php/pid954154?) to report discrimination, hate, or bias incidents. You may report incidents that happen to you anonymously. Please note that, if you report anonymously, the University's ability to respond may be limited.

Class website: Course information and content (including this syllabus, homework assignments, access to WebAssign, recorded lectures, practice material, and much more) will be available at the Blackboard website: <http://my.umbc.edu/go/blackboard>

Approximate Calendar:

Dates	Topics	Reading
8/25 - 8/27	Combinatorial Proofs and Basic Counting	2.1 - 2.4, 3.1 - 3.3, 4.2
9/1 - 9/3	Pigeonhole Principle (PHP)	10.1
9/8 - 9/10	Inclusion-Exclusion and Derangements	10.2, 9.1
9/15 - 9/17	Generating Functions I	7.1 - 7.2
9/22 - 9/24	Generating Functions II	7.3, 9.3
9/29 - 10/1	Linear Recurrences and Catalan Numbers	6.1 - 6.3, 9.2
10/6 - 10/8	Partition Theory and Review	8.1 - 8.2
10/13 - 10/15	No Classes Tuesday 10/13	Thursday 10/15: Midterm Exam #1
10/20 - 10/22	Graph Fundamentals and Adjacency Matrices	11.1 - 11.4
10/27 - 10/29	Trees, Spanning Tress, and Isomorphisms	12.1 - 12.3
11/3 - 11/5	Connectivity and Transversability	12.4, 13.1 - 13.2
11/10 - 11/12	Planarity and Topological Graph Theory	15.1 - 15.3
11/17 - 11/19	Graph Colorings and Matchings	14.1, 14.3, 16.3
11/24 - 11/26	Tuesday 11/24: Midterm Exam #2	No Classes Thursday 11/26
12/1 - 12/3	Ramsey Theory (Extremal Combinatorics)	14.2
12/8	Course Synthesis	
12/17	Thursday 12/17, 8–10am: Final Exam	